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MODULAR ADAPTER ASSEMBLY

Abstract

A modular electrical adapter assembly for connecting a plurality of flat cables and a plurality of wires may include a plurality of male connectors, female connectors, and adapter modules. The adapter modules may be releasably connected to one another. Each adapter module may include a first housing shell, a second housing shell, an interior space defined between the housing shells, a wire opening via which an associated wire extends into the interior space, a cable opening via with an associated flat cable extends into the interior space, a male connector disposed on the second housing shell outside of the interior space, and a female connector disposed on the first housing shell outside of the interior space. Adjacent adapter modules may be releasably connected to one another via engagement of the male connector of a first adapter module and the female connector of an adjacent second adapter module.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to modular adapter assemblies for connecting one or more conductor members together, such as adapter assemblies that may be utilized in connection with and/or incorporated into vehicles for example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and embodiments are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

[0003] FIG. **1** is a perspective view generally illustrating an embodiment of a modular adapter assembly according to teachings of the present disclosure.

[0004] FIG. **2** is a partial perspective view generally illustrating an embodiment of a flat cable according to teachings of the present disclosure.

[0005] FIGS. **3**A and **3**B are partial perspective views generally illustrating an embodiment of a wire including a wire terminal according to teachings of the present disclosure.

[0006] FIGS. **4** and **5** are perspective views generally illustrating an embodiment of an adapter module according to teachings of the present disclosure.

[0007] FIGS. **6**A and **6**B are perspective views generally illustrating an embodiment of a first housing shell of an adapter module according to teachings of the present disclosure.

[0008] FIGS. **7**A and **7**B are perspective views generally illustrating an embodiment of a second housing shell of an adapter module according to teachings of the present disclosure.

[0009] FIGS. **8**, **9**, and **10** are cross-sectional perspective views generally illustrating an embodiment of an adapter module engaging a plurality of wires and a flat cable according to teachings of the present disclosure.

DETAILED DESCRIPTION

[0010] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0011] Referring to FIG. **1**, a modular electrical adapter assembly **100** for connecting a plurality of conductor members **10** (e.g., a plurality of flat cables **20**.sub.1, **20**.sub.2 and a plurality of non-flat wires **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2) is depicted. The adapter assembly **100** (i) accommodates, receives, supports, and/or is connected to the conductor members **10**, (ii) facilitates and/or guides connecting of the conductor members **10** to one another, (iii) enhances, reinforces and/or strengthens a connection of the conductor members **10** to one another and (iv), optionally, secures and/or attaches the conductor members **10** to one or more objects and/or structures. The adapter assembly **100** includes a plurality of adapter modules **110**.sub.1, **110**.sub.2, a plurality of male connectors **200**.sub.1-**200**.sub.3 and a plurality of female connectors **300**.sub.1, **300**.sub.2. Optionally, the adapter assembly **100** also includes a mount **400** via which the adapter assembly **100** and/or the adapter modules **110**.sub.1, **110**.sub.2 are securable to a structure (e.g., one or more portions of a vehicle, such as a vehicle floor and/or panel). The adapter modules **110**.sub.1, **110**.sub.2 and the mount **400** are disposed on top of one another (e.g., in a stacking direction) in a stacked arrangement. The adapter modules **110**.sub.1, **110**.sub.2 and the mount **400** are releasably connected to one another by the male and female connectors **200**.sub.1-**200**.sub.3, **300**.sub.1, **300**.sub.2 via adjusting and/or moving one or more of the adapter modules **110**.sub.1, **110**.sub.2 and/or the mount **400** in a connection direction, which extends perpendicular to the stacking direction.

[0012] The male connectors **200**.sub.1-**200**.sub.3 are each configured in a complimentary manner to the female connectors **300**.sub.1, **300**.sub.2 and releasably engage one or more female connectors **300**.sub.1, **300**.sub.2 (e.g., of an adjacent adapter module **110**.sub.1, **110**.sub.2 and/or mount **400**). The female connectors **300**.sub.1, **300**.sub.2 are each configured in a complimentary manner to the male connectors **200**.sub.1-**200**.sub.3 and releasably engage one or more male connectors **200**.sub.1-**200**.sub.3 (e.g., of an adjacent adapter module **110**.sub.1, **110**.sub.2 and/or mount **400**). Each adapter module **110**.sub.1, **110**.sub.2 includes at least one of the male connectors **200**.sub.1-**200**.sub.3 and one of the female connectors **300**.sub.1, **300**.sub.2. The mount **400** also includes at least one of the male connectors **200**.sub.1-**200**.sub.3, at least one of the female connectors **300**.sub.1, **300**.sub.2, and/or another type of connector that is releasably engageable with the male connectors **200**.sub.1-**200**.sub.3 and/or the female connectors **300**.sub.1, **300**.sub.2.

[0013] Each adapter module **110**.sub.1, **110**.sub.2 is connectable to one or more adjacent adapter modules **110**.sub.1, **110**.sub.2 and/or mounts **400** via (i) engagement of its male connector **200**.sub.1-**200**.sub.3 with the female connector **300**.sub.1, **300**.sub.2 of an adjacent adapter module **110**.sub.1, **110**.sub.2 and/or mount **400** and (ii) engagement of its female connector **300**.sub.1, **300**.sub.2 with the male connector **200**.sub.1-**200**.sub.3 of another adjacent adapter module **110**.sub.1, **110**.sub.2 and/or mount **400**. For example, in FIG. **1**, the first adapter module **110**.sub.1 is connected to the adjacent second adapter module **110**.sub.2 and the adjacent mount **400**. The male connector **200**.sub.1 of the first adapter module **110**.sub.1 is engaged with the female connector **300**.sub.2 of the second adapter module **110**.sub.2 thereby releasably connecting the first adapter module **110**.sub.1 (e.g., the first housing shell) and the second adapter module **110**.sub.2 (e.g., the second housing shell). The female connector **300**.sub.1 of the first adapter module **110**.sub.1 is engaged with the male connector **200**.sub.3 of the mount **400** thereby releasably connecting the first adapter module **110**.sub.1 (e.g., the second housing shell) and the mount **400**. In this way, the first adapter module **110**.sub.1 indirectly connects the second adapter module **110**.sub.2 and the mount **400** to one another. The male connector **200**.sub.2 of the second adapter module **110**.sub.2 is not currently engaged, but could be engaged the female connector of another undepicted adapter module and/or mount to secure the undepicted adapter module and/or mount thereto and, in this way, expand the size and/or capacity of the module adapter assembly **100**.

[0014] The adapter modules **110**.sub.1, **110**.sub.2, male connectors **200**.sub.1-**200**.sub.3, female connectors **300**.sub.1, **300**.sub.2, flat cables **20**.sub.1, **20**.sub.2, and wires **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2 of the exemplary modular adapter assembly **100** depicted in FIG. **1** are substantially similar and/or identical to one another. As such, for brevity, a single adapter module **110** that includes a male connector **200** and a female connector **300**, and that connects a flat cable **20** to a plurality of non-flat wires **40**A-**40**D is depicted in FIGS. **4-10** and described below. The

adapter module **110**, male connector **200**, female connector **300**, flat cable **20**, and wires **40**A-**40**D are representative of the adapter modules **110**.sub.1, **110**.sub.2, male connectors **200**.sub.1-**200**.sub.3, female connectors **300**.sub.1, **300**.sub.2, flat cables **20**.sub.1, **20**.sub.2, and wires **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2 depicted in FIG. **1**, which are therefore not described in detail for brevity. Nevertheless, in other embodiments, an adapter module **110**.sub.1, **110**.sub.2, male connector **200**.sub.1-**200**.sub.3, female connector **300**.sub.1, **300**.sub.2, flat cable **20**.sub.1, **20**.sub.2, and/or wire **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2 may be configured differently than one or more other adapter modules **110**.sub.1, **110**.sub.2, male connectors **200**.sub.1-**200**.sub.3, female connectors **300**.sub.1, **300**.sub.2, flat cables **20**.sub.1, **20**.sub.2, and/or wires **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2 of the modular adapter module **100**, respectively.
[0015] The flat cable **20** depicted in FIGS. **2** and **8-10** is representative of each of the flat cables **20**.sub.1, **20**.sub.2 of FIG. **1**. As generally illustrated in FIG. **2**, the flat cable **20** is flexible and has a substantially rectangular cross-section, though other suitable configurations (e.g., nonflexible) and/or cross-sectional shapes (e.g., polygonal, trapezoidal, octagonal) are possible. The flat cable **20** includes one or more electrical conductors **22**A-**22**D overmolded and/or embedded within an insulating material **24** that electrically insulates the conductors **22**A-**22**D from each other and/or from other components. Optionally, the flat cable **20** also includes a reinforcement body **26** that is connected to and/or embedded within the insulating material **24**. The conductors **22**A-**22**D are disposed substantially parallel with each other and/or substantially aligned in a common plane. The flat cable **20** also includes a plurality of slots **28**A-**28**C configured to receive a respective retention protrusion **142**A-**142**C of the first housing shell **130**. The slots **28**A-**28**C are each disposed in and/or defined by the insulating material **124** and the reinforcement body **126**, such as between adjacent conductors **22**A-**22**D. The slots **28**A-**28**C receive and engage the retention protrusions **142**A-**142**C of the first housing shell **130** (see FIGS. **8** and **10**), which connects the flat cable **20** to the first housing shell **130** and/or the adapter module **110** (to at least an extent) and restricts and/or prevents removal of the flat cable **20** from the cable opening **114**. The flat cable **20** further includes a plurality of terminal recesses **30**A-**30**D that are each configured to receive a nub **58** of a wire terminal **46** of a respective wire **40**A-**40**D. The terminal recesses **30**A-**30**D are each disposed in and/or defined by an exposed portion of a respective conductor **22**A-**22**D. The terminal recesses **30**A-**30**D receive and engage the nubs **58** of the wire terminals **46** of the wires **40**A-**40**D (see FIG. **10**) establishing an electrical connection between the wires **40**A-**40**D and the flat cable **20**, while also restricting and/or resisting movement of the wire terminals **46** relative to the conductors **22**A-**22**D. As a result, a more secure and stable physical and electrical connection is established between (i) each wire terminal **26** and the respective conductor **22**A-**22**D and (ii) the wires **40**A-**40**D and the flat cable **20**.
[0016] The wire **40** depicted in FIGS. **3**A and **3**B is representative of each of the wires **40**A.sub.1-**40**D.sub.1, **40**A.sub.2-**40**D.sub.2 of FIG. **1** and the wires **40**A-**40**D of FIGS. **8-10**. As generally illustrated in FIGS. **3**A and **3**B, the wire **40** is flexible and has a round and/or substantially circular cross-section, though other suitable configurations (e.g., nonflexible) and/or cross-sectional shapes (e.g., polygonal, trapezoidal, octagonal) are possible. The wire **40** includes an electrical conductor **42** overmolded and/or embedded within an insulating material **44** that electrically insulates the conductor **42** from other components. The wire **40** also includes a wire terminal **46** that is connected to the insulating material **44** and/or the conductor **42**. The wire terminal **46** includes a terminal body **48**, a protuberance **56**, and a nub **58**. The terminal body **48** includes a latching portion **50**, a contact portion **52**, and a biasing portion **54**. The protuberance **56** is disposed on and projects transversely from the latching portion **50** of the terminal body **48** (e.g., in the stacking direction). The protuberance **56** is received in and engages the depression **148**A-**148**D of a respective terminal support **146**A-**146**D of the first housing shell **130** (see FIGS. **9** and **10**) and, to at least an extent, restricts movement of the wire terminal **46** relative to the respective terminal support **146**A-**146**D. The nub **58** is disposed on and projects transversely from the contact portion

52 of the terminal body **48** (e.g., in the stacking direction). The nub **58** is received in and engages the terminal recess **30**A-**30**D of a respective conductor **22**A-**22**D of the flat cable **20** (see FIGS. **8** and **10**) establishing an electrical connection therebetween, while also restricting movement of the wire terminal **46** relative to the respective conductor **22**A-**22**D to at least an extent. The protuberance **56** and nub **58** are each structured as embossments of the terminal body **48** and, thus, are formed and defined by portions of the terminal body **48**. Nevertheless, the protuberance **56** and/or nub **58** may alternatively be configured and/or structured as a protrusion, projection, extension, or other type of suitable body. The latching portion **50** is disposed at a fixed end of the wire terminal **46** and is connected to the conductor **42**. The latching portion **50** is structured in a complimentary manner to a respective pair of latches **152**A-**152**D, **154**A-**154**D of the first housing shell **130** and is engaged by the respective pair of latches **152**A-**152**D, **154**A-**154**D to secure the wire **40** and the wire terminal **46** to a respective terminal support **146**A-**146**D, the first housing shell **130**, and/or the adapter module **110** (see FIGS. **8** and **9**). The contact portion **52** extends between and connects the latching portion **50** and the biasing portion **54**. The contact portion **52** is offset from the latching portion **50** and the biasing portion **54** in a direction toward the first housing shell **130** (e.g., in the stacking direction), and contacts, abuts, rests on, and/or engages a respective conductor **22**A-**22**D of the flat cable **20** (see FIGS. **8** and **10**). The biasing portion **54** is disposed at a free end of the terminal body **48**. The biasing portion **54** is contacted and/or pressed on by the cable rib **190** and/or the lip **194** (e.g., toward the first housing shell **130** and/or the respective conductor **22**A-**22**D in the stacking direction) thereby (i) biasing the contact portion **52** of the wire terminal **46** against the respective conductor **22**A-**22**D, (ii) restricting and/or preventing movement of the wire terminal **46** away from the respective conductor **22**A-**22**D, and/or (iii) restricting and/or preventing removal of the nub **58** from the terminal recess **30**A-**30**D of the respective conductor **22**A-**22**D (see FIGS. **8** and **10**).

[0017] The adapter module **110** depicted in FIGS. **4-10** is representative of each adapter module **110**.sub.1, **110**.sub.2 of FIG. **1**. As generally shown in FIGS. **8-10**, the adapter module (i) accommodates, receives, supports, and/or is connected to a respective group and/or subset of conductor members **10** (e.g., flat cable **20** and wires **40**A-**40**D), (ii) facilitates and/or guides connecting of the flat cable **20** and the wires **40**A-**40**D together, and (iii) enhances, reinforces and/or strengthens a connection of the flat cable **20** and the wires **40**A-**40**D to one another. An adapter module **110**, **110**.sub.1, **110**.sub.2 may additionally and/or alternatively be considered, referred to, and/or claimed as an adapter, an electrical adapter, an adapter unit, a housing, and/or a connection unit, among other suitable and appropriate nomenclatures.

[0018] As generally illustrated in FIGS. **4-8**, the adapter module **110** includes an interior space **112**, a cable opening **114**, a plurality of wire openings **116**A-**116**D, a first housing shell **130**, and a second housing shell **170**. The housing shells **130**, **170** are separate and independent components and/or bodies. The housing shells **130**, **170** are composed of a plastic material, but may additionally and/or alternatively be composed of and/or include one or more other materials such as a metal, rubber, ceramic, and/or composite. The housing shells **130**, **170** are releasably and/or removably connectable to one another via complimentary fasteners **136**A, **136**B, **176**A, **176**B. The internal space **112**, the cable opening **114**, and the wire openings **116**A-**116**D are defined by and between the housing shells **130**, **170**. The cable opening **114** removably receives at least a portion of the flat cable **20** and/or the flat cable **20** extends into the internal space **112** via the cable opening **114**. The wire openings **116**A-**116**D each removably receive at least a portion of a respective wire **40**A-**40**D and/or the wires **40**A-**40**D each extend into the internal space **112** via a respective wire opening **116**A-**116**D. The adapter module **110** may be considered and/or referred to being in an assembled configuration and/or a use configuration when the housing shells **130**, **170** are connected and the adapter module **110** is engaged and/or connected to the respective flat cable **20** and the respective wires **40**A-**40**D (i.e., the respective flat cable **20** is disposed in the cable opening **114** and the respective wires **40**A-**40**D are disposed in the wire openings **116**A-**116**D).

[0019] As generally illustrated in FIGS. **4-6**A, **7**A, **9**, and **10**, the adapter module **110** includes one of the male connectors **200** and one of the female connectors **300**. The male and female connector **200**, **300** project outward from the adapter module **110** and/or the housing shells **130**, **170** and are disposed outside of the interior space **112** of the adapter module **110**. The male connector **300** is disposed on and/or connected to one of the housing shells **130**, **170** (e.g., the second housing shell **170**) and the female connector **300** is disposed on and/or connected to the other one of the housing shells **130**, **170** (e.g., the first housing shell **130**). In other words, the first housing shell **130** includes one of the male connector **200** and the female connector **300** and the second housing shell **170** includes the other one of the male connector **200** and the female connector **300**. Therefore, the first housing shell **130** of the adapter module **110** is releasably connectable to the second housing shell **170** of one or more other adapter modules **110** and/or to a mount **400** that includes a complimentary connector (e.g., one of the female connectors **300**). The second housing shell **170** of the adapter module **110** is releasably connectable to the first housing shell **130** of one or more other adapter modules **110** and/or to a mount **400** that includes a complimentary connector (e.g., one of the male connectors **200**).
[0020] As generally illustrated in FIGS. **4**, **5**, **7**A, **9**, and **10**, the male connector **200** is configured in a complimentary manner to one or more female connectors **300** (e.g., the female connector **300**.sub.1, **300**.sub.2 of each adapter module **110**.sub.1, **110**.sub.2 and/or mount **400** of the assembly **100**). The male connector **200** adjustably (e.g., slidably) and releasably engages a female connector **300** to connect the adapter module **110** to another adapter module **110** and/or the mount **400**. At least a portion of the male connector **200** is insertable into and removable from the female connector **300** (e.g., the attachment receptacle **306**) in a connection direction. Additionally, the male connector **200** receives and releasably engages a portion of the female connector **300** (e.g., the tab **302**). The male connector **200** includes a main portion **202**, a pair of guide rails **204**A, **204**B, and an attachment flange **206**. The main portion **202** is disposed on, connected to, and/or projects outwardly from the base **172** of the second housing shell **170** (e.g., in the stacking direction). The attachment flange **206** is disposed at the free end of the main portion **202** and is disposed spaced apart from the base **172** of the second housing shell **170**. The attachment flange **206** projects from the main portion **202** generally toward the wire openings **116**A-**116**D and/or the wire apertures **182**A-**182**D (e.g., in the connection direction). The attachment flange **206** includes and defines a flange recess **208** that receives and engages the tab **302** of the female connector **300**. The attachment flange **206** is configured to elastically deform and/or flex away from the second housing shell **170** (e.g., in the stacking direction) when engaging and disengaging the tab **302** of the female connector **300**. The pair of guide rails **204**A, **204**B are slidably engageable with the pair of tracks **304**A, **304**B of the female connector **300**. The pair of guide rails **204**A, **204**B includes a first lateral protrusion and/or rail **204**A and a second lateral protrusion and/or rail **204**B. The lateral protrusions **204**A, **204**B are disposed at a free end of the main portion **202** and are disposed spaced apart from the base **172** of the second housing shell **170**. The lateral protrusions **204**A, **204**B protrude laterally (e.g., transversely and/or perpendicularly) from opposite sides of the main portion **202** (e.g., transversely and/or perpendicularly to the stacking direction and the connection direction) such that the main portion **202** and the lateral projections **204**A, **204**B form and/or define a T-shaped cross-sectional profile. The lateral protrusions **204**A, **204**B each adjustably (e.g., slidably) engage a respective L-shaped projection **304**A, **304**B of the female connector **300**. The lateral protrusions **204**A, **204**B, together with the L-shaped projections **304**A, **304**B, guide the attachment flange **206** into engagement with the tab **302** when engaging the male and female connector **200**, **300**.
[0021] As generally illustrated in FIGS. **4**, **5**, **6**A, **9**, and **10**, the female connector **300** is configured in a complimentary manner to one or more male connectors **200** (e.g., the male connector **200**.sub.1-**200**.sub.3 of each adapter module **110**.sub.1, **110**.sub.2 and/or mount **400** of the assembly **100**). The female connector **300** adjustably (e.g., slidably) and releasably receives and/or engages a male connector **200** to connect the adapter module **110** to another adapter module **110**

and/or the mount **400**. At least a portion of the male connector **200** is insertable into and removable from the female connector **300** in the connection direction. Additionally, at least a portion of the female connector **300** is received in and releasably engaged by the male connector **200**. The female connector **300** includes a tab **302** and a pair of tracks **304**A, **304**B. The pair of tracks **304**A, **304**B includes and/or is defined by one or more opposing L-shaped projections (e.g., a first L-shaped projection and/or track **304**A, a second L-shaped projection and/or track **304**B. The L-shaped projections **304**A, **304**B and the tab **302** are disposed on, connected to, and/or project outwardly from the base **132** of the first housing shell **130** (e.g., in the stacking direction). The L-shaped projections **304**A, **304**B collectively define an attachment receptacle **306** that receives at least a portion of the male connector **200**. The L-shaped projections **304**A, **304**B face one another and each receive and adjustably (e.g., slidably) engage a respective lateral protrusion **204**A, **204**B of the male connector **200**. The tab **302** is disposed adjacent to the L-shaped projections **304**A, **304**B and is configured to be received in and engage (e.g., snap into engagement with) the attachment flange **206** of the male connector **200**. The tab **302** is disposed in alignment with the attachment receptacle **306** (e.g., in the connection direction) and/or, in other non-depicted embodiments, projects into the attachment receptacle **306**.

[0022] As generally illustrated in FIGS. **4-6**B, and **8-10**, the first housing shell **130** is releasably connectable to the second housing shell **170**. The first housing shell **130** includes a base **132** and one or more sidewalls (e.g., a first, second, and third sidewall **134**A-**134**C). The sidewalls **134**A-**134**C project from the base **132** (e.g., in the stacking direction) and extend at least partially around an outer perimeter of the base **132**. The first housing shell **130** further includes a plurality of first fasteners **136**A, **136**B that are connected to, disposed on, and/or protrude from the base **132** and/or a respective one of the second and third sidewalls **134**B, **134**C (e.g., in the stacking direction). The first fasteners **136**A, **136**B are configured in a complimentary manner to second fasteners **176**A, **176**B of the second housing shell **170** and releasably engage the second fasteners **176**A, **176**B to releasably connect the housing shells **130**, **170** together.

[0023] As generally illustrated in FIGS. **6**B, **8**, and **9**, the first housing shell **130** includes a plurality of support walls **140**A-**140**C. The support walls **140**A-**140**C are disposed on and project transversely (e.g., perpendicularly) from the base **132** toward the second housing shell **170** (e.g., in the stacking direction). The support walls **140**A-**140**C extend longitudinally along the base **132** transversely (e.g., perpendicularly) to the first sidewall **134**A and/or substantially parallel to the axis of the cable opening **114** (e.g., in the connection direction). The support walls **140**A-**140**C are each disposed in alignment with a respective retention protrusion **142**A-**142**C (e.g., in the connection direction) and/or a respective support wall **180**A-**180**C of the second housing shell **170** (e.g., in the stacking direction) at least when the adapter module **110** is in the assembled configuration. The support walls **140**A-**140**C are each disposed between a respective pair of adjacent terminal supports **146**A-**146**D and/or a respective pair of latches **152**A-**152**D, **154**A-**154**D (e.g., the first latch **152**A-**152**D of one latch pair and the second latch **154**A-**154**D of another adjacent latch pair). When the housing shells **130**, **170** are connected, the support walls **140**A-**140**C each contact, abut, and/or rest on a respective support wall **180**A-**180**C of the second housing shell **170** (e.g., in the stacking direction). In this manner, the support walls **140**A-**140**C contact and support the second housing shell **170** and increase the structural integrity of the adapter module **110** (e.g., resist and/or mitigate deformation and/or flexing of the bases **132**, **172** of the housing shells **130**, **170**; resist and/or prevent crushing of the adapter module **110**).

[0024] As generally illustrated in FIGS. **6**B, **8**, and **10**, the first housing shell **130** includes a plurality of retention protrusions **142**A-**142**C. The retention protrusions **142**A-**142**C are disposed on and project transversely (e.g., perpendicularly) from the base **132** toward the second housing shell **170** (e.g., in the stacking direction). The retention protrusions **142**A-**142**C extend longitudinally along the base **132** transversely (e.g., perpendicularly) to the first sidewall **134**A and/or substantially parallel to an axis of the cable opening **114** (e.g., the connection direction). The retention protrusions **142**A-**142**C are each disposed in alignment with a respective slot **28**A-**28**C of the flat cable **20** (e.g., in the stacking direction) and/or a respective support wall **140**A-**140**C (e.g., in the connection direction) at least when the adapter module **110** is in the assembled configuration. When the adapter module **110** is in the assembled configuration, the retention protrusions **142**A-**142**C are each disposed in and/or engage a respective slot **28**A-**28**C of the flat cable **20**, which connects and/or secures the flat cable **20** to the first housing shell **130** and/or the adapter module **110** (to at least an extent) and restricts and/or prevents removal of the flat cable **20** from the cable opening **114**.

[0025] As generally illustrated in FIGS. **6**B, **8**, and **9**, the first housing shell **130** includes a plurality of terminal supports **146**A-**146**D that each support the wire terminal **46** of a respective wire **40**A-**40**D. The terminal supports **146**A-**146**D are disposed on and project transversely (e.g., perpendicularly) from the base **132** toward the second housing shell **170** (e.g., in the stacking direction). The terminal supports **146**A-**146**D are each disposed in alignment with a respective wire aperture **182**A-**182**D and a respective terminal rib **186**A-**186**D (e.g., in the connection direction) when the housing shells **130**, **170** are connected. The terminal supports **146**A-**146**D extend longitudinally along the base **132** substantially parallel to an axis of the respective wire aperture **182**A-**182**D and/or an axis of the cable opening **114** (e.g., the connection direction) and/or transversely (e.g., perpendicularly) to the first sidewall **134**A. When the adapter module **110** is in the assembled configuration, the wire terminal **46** of each wire **40**A-**40**D is disposed on and supported by a respective terminal support **146**A-**146**D, and are secured on the respective terminal support **146**A-**146**D by a respective pair of latches **152**A-**152**D, **154**A-**154**D. Each terminal support **146**A-**146**D includes and/or defines a depression **148**A-**148**D that receives and engages the protuberance **56** of the wire terminal **46** of the respective wire **40**A-**40**D, which, to at least an extent, restricts movement of the respective wire terminal **46** relative to the terminal support **146**A-**146**D, the first housing shell **130**, and/or the adapter module **110**. The depression **148**A-**148**D opens towards the second housing part **170** (e.g., in the stacking direction) and is at least partially aligned with a respective terminal rib **186**A-**186**D (e.g., in the stacking direction) at least when the adapter module **110** is in the assembled configuration.

[0026] As generally illustrated in FIGS. **6**B, **8**, and **9**, the first housing shell **130** includes a plurality of latches **152**A-**152**D, **154**A-**154**D configured to engage the wire terminals **46** of the wires **40**A-**40**D. The latches **152**A-**152**D, **154**A-**154**D are disposed on and project transversely (e.g., perpendicularly) from the base **132** toward the second housing shell **170** (e.g., in the stacking direction). The latches **152**A-**152**D, **154**A-**154**D each include a latching nose disposed at or about and projecting from a free end of the respective latch **152**A-**152**D, **154**A-**154**D. The plurality of latches **152**A-**152**D, **154**A-**154**D includes a plurality of latch pairs (e.g., a first latch pair **152**A, **154**A; a second latch pair **152**B, **154**B; etc.) that each contact and/or engage the wire terminal **46** of a respective wire **40**A-**40**D and connect, fasten, and/or secure the respective wire terminal **46** to and/or on a respective terminal support **146**A-**146**D, the first housing shell **130**, and/or the adapter module **110**. Each latch pair includes a first latch **152**A-**152**D and a second latch **154**A-**154**D that simultaneously contact and/or engage the respective wire terminal **46** (e.g., via their latching noses). The first and second latch **152**A-**152**D, **154**A-**154**D of each latch pair are disposed on opposite sides of the respective terminal support **146**A-**146**D and face toward one another. The first latches **152**A-**152**D and the second latches **154**A-**154**D are substantially mirror images of one another, but may alternatively be configured differently than one another in other embodiments. When connecting a wire **40**A-**40**D to the first housing shell **130** and/or the adapter module **110**, the wire terminal **46** of the wire **40**A-**40**D is pressed against the sloped surfaces of the latching noses of the respective latching pair (e.g., in the stacking direction and/or toward the terminal support **146**A-**146**D), which causes the first and second latches **152**A-**152**D, **154**A-**154**D of the latching pair to elastically deform and/or flex away from one another (e.g., transversely to the connection direction and the stacking direction) to allow the wire terminal **46** to be disposed on the respective terminal support **146**A-**146**D.

[0027] As generally illustrated in FIGS. **4**, **5**, **7**A, **7**B, and **8-10**, the second housing shell **170** is releasably connectable to the first housing shell **130**. The second housing shell **170** includes a base **172** and one or more sidewalls (e.g., a first, second, third, and fourth sidewall **174**A-**174**D). The sidewalls **174**A-**174**D project from the base **172** (e.g., in the stacking direction) and extend at least partially around an outer perimeter of the base **172**. The second housing shell **170** further includes a plurality of second fasteners **176**A, **176**B that are connected to, disposed on, and/or protrude from the base **172** and/or a respective one of the second and third sidewalls **174**B, **174**C (e.g., in the stacking direction). The second fasteners **176**A, **176**B are configured in a complimentary manner to the first fasteners **136**A, **136**B of the first housing shell **130** and releasably engage the first fasteners **136**A, **136**B to releasably connect the housing shells **130**, **170** together.

[0028] As generally illustrated in FIGS. **7**B, **8**, and **9**, the second housing shell **170** includes a plurality of support walls **180**A-**180**C. The support walls **180**A-**180**C are disposed on and project transversely (e.g., perpendicularly) from the base **172** toward the first housing shell **130** (e.g., in the stacking direction). The support walls **180**A-**180**C extend longitudinally along the base **172** transversely (e.g., perpendicularly) to the first and fourth sidewalls **174**A, **174**D and/or substantially parallel to the axis of the cable opening **114** (e.g., in the connection direction). The support walls **180**A-**180**C are each disposed in alignment with a respective support wall **140**A-**140**C of the first housing shell **130** (e.g., in the stacking direction) at least when the housing shells **130**, **170** are connected. The support walls **180**A-**180**C are each disposed between a respective pair of adjacent terminal ribs **186**A-**186**D and/or a respective pair of adjacent wire apertures **182**A-**182**D. The support walls **180**A-**180**C extend between and connect the respective pair of adjacent terminal ribs **186**A-**186**D, which enhances the strength and rigidity of the terminal ribs **186**A-**186**D to at least an extent. When the housing shells **130**, **170** are connected, the support walls **180**A-**180**C each contact, abut, and/or rest on a respective support wall **140**A-**140**C of the first housing shell **130** (e.g., in the stacking direction). In this manner, the support walls **180**A-**180**C contact and support the first housing shell **130** and increase the structural integrity of the adapter module **110** (e.g., resist and/or mitigate deformation and/or flexing of the bases **132**, **172** of the housing shells **130**, **170**; resist and/or prevent crushing of the adapter module **110**).

[0029] As generally illustrated in FIGS. **7**B, **8**, and **9**, the second housing shell **170** includes a plurality of wire apertures **182**A-**182**D that are disposed in and defined by the first sidewall **174**A. The wire apertures **182**A-**182**D project into the first sidewall **174**A toward the base **172** (e.g., in the stacking direction) from a free end of the first sidewall **174**A. When the housing shells **130**, **170** are connected, the free end of the first sidewall **174**A contacts and/or is disposed adjacent to free end of the first sidewall **134**A of the first housing shell **130** such that the first sidewall **134**A and/or the first housing shell **130** closes the wire apertures **182**A-**182**D to form the wire openings **116**A-**116**D (i.e., the wire openings **116**A-**116**D are defined by and between the first sidewalls **134**A, **174**A of the housing shells **130**, **170**).

[0030] As generally illustrated in FIGS. **5**, **7**B, and **8**, the fourth sidewall **174**D of the second housing shell **170** is disposed opposite the first sidewall **174**A and at least partially defines the cable opening **114**. When the housing shells **130**, **170** are connected, the fourth sidewall **174**D contacts, presses against, and/or is disposed in close proximity to the flat cable **20** such that, for example, the flat cable **20** is disposed and/or sandwiched between the fourth sidewall **174**D and the base **132** of the first housing shell **130**.

[0031] As generally illustrated in FIGS. **7**B, **8**, and **9**, the second housing shell **170** includes a plurality of terminal ribs **186**A-**186**D that restrict disengagement of the latches **152**A-**152**D, **154**A-**154**D and the wire terminals **46** of the wires **40**A-**40**D. The terminals ribs **186**A-**186**D are disposed on and project transversely (e.g., perpendicularly) from the base **172** toward the first housing shell **130** (e.g., in the stacking direction). The terminal ribs **186**A-**186**D are each disposed in alignment with a respective wire aperture **182**A-**182**D and a respective terminal support **146**A-**146**D (e.g., in the connection direction) at least when the adapter module **110** is in the assembled configuration. The terminals ribs **186**A-**186**D extend longitudinally along the base **172** transversely (e.g., perpendicularly) to an axis of the respective wire aperture **182**A-**182**D and/or an axis of the cable opening **114** (e.g., the connection direction) and/or substantially parallel to the first and fourth sidewalls **174**A, **174**D. Each terminal rib **186**A-**186**D includes notches **188**A-**188**D that receive (e.g., in the stacking direction) and engage at least a portion of a respective pair of latches **152**A-**152**D, **154**A-**154**D of the first housing shell **130** when the housing shells **130**, **170** are connected, which restricts and/or prevents the latches **152**A-**152**D, **154**A-**154**D from adjusting and/or flexing away from one another (e.g., transversely to the connection direction and the stacking direction) and/or from disengaging the respective wire terminal **46**. Each terminal rib **186**A-**186**D has a free end that is at least partially disposed between the respective pair of latches **152**A-**152**D, **154**A-**154**D and that contacts and/or is disposed in close proximity to the respective wire terminal **46** (e.g., the latching portion **50** thereof) when the adapter module **110** is in the assembled configuration, which (i) restricts movement of the respective wire terminal **46** relative to the terminal support **146**A-**146**D to at least an extent and (ii) restricts and/or prevents removal of the protuberance **56** of the respective wire terminal **46** from the depression **148**A-**148**D of the terminal support **146**A-**146**D. In addition, if one or more of the wire terminals **46** is not properly seated on the respective terminal support **146**A-**146**D and/or is not properly engaged with the respective latches **152**A-**152**D, **154**A-**154**D when attempting to connect the housing shells **130**, **170**, the corresponding terminal rib **186**A-**186**D contacts and/or abuts the improperly seated and/or engaged wire terminal **46** and presses it against the associated terminal support **146**A-**146**D, which effectively prevents and/or blocks further adjustment and/or movement of the housing shells **130**, **170** in the stacking direction. This in turn prevents and/or blocks the first and second fasteners **136**A, **136**B, **176**A, **176**B from being adjusted and/or moved into engagement with one another and, thus, prevents the housing shells **130**, **170** from being connected. The terminal ribs **186**A-**186**D (e.g., in conjunction with the terminal supports **146**A-**146**D) therefore also serve a position and connection assurance function with respect to the wires **40**A-**40**D and/or wire terminals **146**A-**146**D and the latches **152**A-**152**D, **154**A-**154**D of the first housing shell **130**.

[0032] As generally illustrated in FIGS. **7**B, **8**, and **10**, the second housing shell **170** includes a cable rib **190**. The cable rib **190** is disposed on and projects transversely (e.g., perpendicularly) from the base **172** toward the first housing shell **130**. The cable rib **190** extends longitudinally along the base **172** transversely (e.g., perpendicularly) to the axis of the cable opening **114** and/or substantially parallel to the first and fourth sidewalls **174**A, **174**D. A free end of the cable rib **190** and/or one or more elevations **192**A-**192**D projecting therefrom contact, press against, and/or are disposed in close proximity to the flat cable **20** (e.g., a respective conductor **22**A-**22**D thereof) when the adapter module **110** is in the assembled configuration such that, for example, the flat cable **20** is disposed and/or sandwiched between the free end of the cable rib **190** (e.g., the elevations **192**A-**192**D) and the base **132** of the first housing shell **130**. In this way, the cable rib **190** (i) helps to maintain a position of the flat cable **20** in the adapter module **110**, (ii) restricts movement of the flat cable **20** relative to the adapter module **110**, the housing shells **130**, **170**, and/or the wire terminals **46** to at least an extent, and (iii) restricts and/or prevents removal of the retention protrusions **142**A-**142**C of the first housing shell **130** from the slots **28**A-**28**C of the flat cable **20**.

[0033] As generally illustrated in FIGS. **7**B, **8**, and **10**, the second housing shell **170** and/or the cable rib **190** includes a lip **194**. The lip **194** is disposed on the cable rib **190** and/or the base **172**, and projects transversely (e.g., perpendicularly) from the cable rib **190** toward the first sidewall **174**A, and/or the wire apertures **182**A-**182**D (e.g., the wire openings **116**A-**116**D). The lip **194** extends longitudinally along the cable rib **190** and/or the base **172** transversely (e.g., perpendicularly) to the axis of the cable opening **114** and/or substantially parallel to the first and fourth sidewalls **174**A, **174**D. When the adapter module **110** is in the assembled configuration, the lip **194** contacts, presses against, and/or is disposed in close proximity to the free ends and/or biasing portions **54** of the wire terminals **46** of wires **40**A-**40**D (e.g., in the stacking direction) thereby (i) biasing the contact portions **52** of the wire terminals **46** of wires **40**A-**40**D against the conductors **22**A-**22**D of the flat cable **20**, (ii) restricting and/or preventing movement of the wire terminals **46** (e.g., the contact portions **52**) of wires **40**A-**40**D away from the conductors **22**A-**22**D, and/or (iii) restricting and/or preventing removal of the nubs **58** of the wire terminals **46** of wires **40**A-**40**D from the terminal recesses **30**A-**30**D of the conductors **22**A-**22**D.

[0034] As generally illustrated in FIG. **1**, the mount **400** includes a flange **402**, a fastener extension **404**, and at least one connector (e.g., a male connector **200**.sub.3). The mount **400** may alternatively include one of the female connectors **300** and/or another connector that is complimentary to the male connector **200** and/or the female connector **300**. The flange **402** has a hollow, conical configuration. The male connector **200**.sub.3 is disposed at and extends from a first, narrower end of the flange **402**. The fastener extension **404** is disposed on an opposite side of the flange **402** relative to the male connector **200**.sub.3 and extends from the flange **402** generally in a direction away from the male connector **200**.sub.3 (e.g., in the stacking direction). The fastener extension **404** is at least partially encircled by the flange **402** and projects from a second, wider end of the flange **402**. The fastener extension **404** is configured to engage a recess disposed in and/or defined by one or more objects and/or structures (e.g., one or more portions of a vehicle, such as a vehicle floor, a panel, a bracket, etc.) to connect the modular adapter assembly **100** to the object and/or structure. The fastener extension **404** includes one or more engagement projections **406** that engage the object and/or structure to connect the mount **400** and/or assembly **100** thereto. The engagement projections **406** have a generally hollow, conical shape and are outwardly angled, sloped, and/or curved toward the flange **402** (e.g., the engagement projections **406** each have a wider end that opens toward the flange **402**). The engagement projections **406** are disposed spaced apart from one another along the length of the fastener extension **404**. When the mount **400** is connected to the object and/or structure, the second, wider end of the flange **402** presses against the object and/or structure (e.g., a surface thereof facing the flange **402**) and biases one or more of the engagement projections **406** of the fastener extension **404** against the object and/or structure. This restricts and/or limits relative movement between the mount **400** (e.g., the assembly **100**) and the object and/or structure, which may limit, reduce, and/or prevent noise generation (e.g., rattling).

[0035] 1. A modular electrical adapter assembly for connecting a plurality of flat cables and a plurality of wires, comprising a plurality of male connectors; a plurality of female connectors complimentary to the plurality of male connectors; and a plurality of adapter modules releasably connected to one another, each adapter module of the plurality of adapter modules including: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which an associated wire of said plurality of wires extends into the interior space; a cable opening via with an associated flat cable of said plurality of flat cables extends into the interior space; a male connector of the plurality of male connectors, the male connector disposed on the second housing shell outside of the interior space; and a female connector of the plurality of female connectors, the female connector disposed on the first housing shell outside of the interior space; wherein adjacent adapter modules of the plurality of adapter modules are releasably connected to one another via engagement of the male connector of a first adapter module of the adjacent adapter modules and the female connector of a second adapter module of the adjacent adapter modules.

[0036] 2. The modular electrical adapter assembly according to embodiment 1, wherein the plurality of adapter modules are disposed one on top of another in a stacked arrangement.

[0037] 3. The modular electrical adapter assembly according to any of the preceding embodiments, further comprising a mount via which the plurality of adapter modules are securable to a structure, wherein: the mount includes (i) a fastener engageable with said structure and (ii) a connector; the connector is one of a male connector of the plurality of male connectors and a female connector of the plurality of female connectors; and the mount is releasably connected to an adapter module of the plurality of adapter modules via engagement of the connector of the mount and one of the male connector and the female connector of the adapter module.

[0038] 4. An electrical adapter for connecting a flat cable and at least one wire, comprising: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which said wire extends into the interior space; a cable opening via with said flat cable extends into the interior space; a male connector disposed on the second housing shell outside of the interior space; and a female connector disposed on the first housing shell outside of the interior space; wherein the male connector and the female connector are releasably engageable with one another such that (i) the second housing shell is releasably connectable to another first housing shell of another electrical adapter and (ii) the first housing shell is releasably connectable to another second housing shell of another electrical adapter.

[0039] 5. The electrical adapter according to embodiment 4, wherein the first housing shell includes a retention protrusion projecting toward the second housing shell, the retention protrusion disposed in a slot of said flat cable and securing said flat cable to the first housing shell.

[0040] 6. The electrical adapter according to any of the preceding embodiments, wherein the first housing shell includes at least one latch projecting toward the second housing shell, the at least one latch engaging a wire terminal of said wire and securing said wire to the first housing shell.

[0041] 7. The electrical adapter according to any of the preceding embodiments, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib engaging the at least one latch and restricting disengagement of the at least one latch and said wire terminal to at least an extent.

[0042] 8. The electrical adapter according to any of the preceding embodiments, wherein the terminal rib includes a notch in which at least a portion of the at least one latch is disposed.

[0043] 9. The electrical adapter according to any of the preceding embodiments, wherein the first housing shell includes a terminal support projecting toward the second housing shell and on which a wire terminal of said wire is disposed.

[0044] 10. The electrical adapter according to any of the preceding embodiments, wherein the first housing shell includes a pair of latches projecting toward the second housing shell and disposed on opposite sides of the terminal support, the pair of latches facing each other and simultaneously engaging said wire terminal and securing said wire terminal on the terminal support.

[0045] 11. The electrical adapter according to any of the preceding embodiments, wherein the terminal support includes a depression receiving a protuberance of said wire terminal and restricting movement of said wire terminal relative to the terminal support to at least an extent.

[0046] 12. The electrical adapter according to any of the preceding embodiments, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to said wire terminal and restricting removal of said protuberance of said wire terminal from the depression of the terminal support.

[0047] 13. The electrical adapter according to any of the preceding embodiments, wherein the depression opens towards the second housing shell and is at least partially aligned with the terminal rib.

[0048] 14. The electrical adapter according to any of the preceding embodiments, wherein the terminal rib and the terminal support prevent the first housing shell and the second housing shell from being connected to one another when said wire terminal is improperly seated on the terminal support.

[0049] 15. The electrical adapter according to any of the preceding embodiments, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to a wire terminal of said wire and restricting movement of said wire terminal relative to the first housing shell to at least an extent.

[0050] 16. The electrical adapter according to any of the preceding embodiments, wherein the second housing shell includes a cable rib projecting toward the first housing shell, the cable rib having a free end contacting and/or disposed in close proximity to said flat cable and restricting movement of said flat cable relative to the first housing shell to at least an extent.

[0051] 17. The electrical adapter according to any of the preceding embodiments, wherein the second housing shell includes a lip projecting toward the wire opening, the lip contacting and/or disposed in close proximity to a free end of a wire terminal of said wire and restricting movement of said wire terminal away from a conductor of said flat cable engaged therewith.

[0052] 18. The electrical adapter according to any of the preceding embodiments, wherein: the female connector includes (i) a pair of tracks and (ii) a tab; and the male connector includes (i) a pair of guide rails slidably engageable with the pair of tracks and (ii) a latch releasably engageable with the tab.

[0053] 19. An electrical adapter for connecting a flat cable and a wire, comprising: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which said wire extends into the interior space; a cable opening via with said flat cable extends into the interior space; a male connector disposed on the second housing shell outside of the interior space; and a female connector disposed on the first housing shell outside of the interior space; wherein the first housing shell includes: a retention protrusion projecting toward the second housing shell, the retention protrusion disposed in a slot of said flat cable and securing said flat cable to the first housing shell; a terminal support projecting toward the second housing shell and on which a wire terminal of said wire is disposed; and at least one latch disposed adjacent to the terminal support and projecting toward the second housing shell, the at least one latch engaging said wire terminal and securing said wire terminal on the terminal support; wherein the second housing shell includes: a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to said wire terminal and restricting disengagement of the at least one latch and said wire terminal to at least an extent; a cable rib projecting toward the first housing shell, the cable rib having a free end contacting and/or disposed in close proximity to said flat cable and restricting removal of the retention protrusion from said slot of said flat cable; and a lip projecting from the cable rib toward the terminal rib, the lip contacting and/or disposed in close proximity to a free end of said wire terminal and restricting movement of said wire terminal away from a conductor of said flat cable engaged therewith; and wherein the male connector and the female connector are releasably engageable with one another such that (i) the second housing shell is releasably connectable to another first housing shell of another electrical adapter and (ii) the first housing shell is releasably connectable to another second housing shell of another electrical adapter.

[0054] 20. A modular adapter assembly for connecting a plurality of flat cables and a plurality of wires, comprising a plurality of electrical adapters according to any of the preceding embodiments, wherein adjacent adapters of the plurality of adapters are releasably connected to one another via engagement of the male connector of a first adapter of the adjacent adapters and the female connector of a second adapter of the adjacent adapters.

[0055] Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

[0056] Reference throughout the specification to "examples, "in examples," "with examples," "various embodiments," "with embodiments," "in embodiments," or "an embodiment," or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases "examples, "in examples," "with examples," "in various embodiments," "with embodiments," "in embodiments," or "an embodiment," or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof.

[0057] It should be understood that references to a single element are not necessarily so limited and may include one or more of such element. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

[0058] "One or more" includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

[0059] It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the various described embodiments. The first element and the second element are both element, but they are not the same element.

[0060] The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including." "comprises," and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0061] Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements, relative movement between elements, direct connections, indirect connections, fixed connections, movable connections, operative connections, indirect contact, and/or direct contact. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. Connections of electrical components, if any, may include mechanical

connections, electrical connections, wired connections, and/or wireless connections, among others. Uses of “e.g.” and “such as” in the specification are to be construed broadly and are used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples.
[0062] While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, it should be understood that such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.
[0063] As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.
[0064] All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

Claims

1. A modular electrical adapter assembly for connecting a plurality of flat cables and a plurality of wires, comprising: a plurality of male connectors; a plurality of female connectors complimentary to the plurality of male connectors; and a plurality of adapter modules releasably connected to one another, each adapter module of the plurality of adapter modules including: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which an associated wire of said plurality of wires extends into the interior space; a cable opening via with an associated flat cable of said plurality of flat cables extends into the interior space; a male connector of the plurality of male connectors, the male connector disposed on the second housing shell outside of the interior space; and a female connector of the plurality of female connectors, the female connector disposed on the first housing shell outside of the interior space; wherein adjacent adapter modules of the plurality of adapter modules are releasably connected to one another via engagement of the male connector of a first adapter module of the adjacent adapter modules and the female connector of a second adapter module of the adjacent adapter modules.
2. The modular electrical adapter assembly of claim 1, wherein the plurality of adapter modules are disposed one on top of another in a stacked arrangement.
3. The modular electrical adapter assembly of claim 1, further comprising a mount via which the plurality of adapter modules are securable to a structure, wherein: the mount includes (i) a fastener engageable with said structure and (ii) a connector; the connector is one of a male connector of the plurality of male connectors and a female connector of the plurality of female connectors; and the mount is releasably connected to an adapter module of the plurality of adapter modules via engagement of the connector of the mount and one of the male connector and the female connector of the adapter module.
4. An electrical adapter for connecting a flat cable and at least one wire, comprising: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which said wire extends into the interior space; a cable opening via with said flat cable extends into the interior space; a male connector disposed on the second housing shell outside of the interior space; and a female connector disposed on the first housing shell outside of the interior space; wherein the male connector and the female connector are releasably engageable with one another

such that (i) the second housing shell is releasably connectable to another first housing shell of another electrical adapter and (ii) the first housing shell is releasably connectable to another second housing shell of another electrical adapter.

5. The electrical adapter of claim 4, wherein the first housing shell includes a retention protrusion projecting toward the second housing shell, the retention protrusion disposed in a slot of said flat cable and securing said flat cable to the first housing shell.

6. The electrical adapter of claim 4, wherein the first housing shell includes at least one latch projecting toward the second housing shell, the at least one latch engaging a wire terminal of said wire and securing said wire to the first housing shell.

7. The electrical adapter of claim 6, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib engaging the at least one latch and restricting disengagement of the at least one latch and said wire terminal to at least an extent.

8. The electrical adapter of claim 7, wherein the terminal rib includes a notch in which at least a portion of the at least one latch is disposed.

9. The electrical adapter of claim 4, wherein the first housing shell includes a terminal support projecting toward the second housing shell and on which a wire terminal of said wire is disposed.

10. The electrical adapter of claim 9, wherein the first housing shell includes a pair of latches projecting toward the second housing shell and disposed on opposite sides of the terminal support, the pair of latches facing each other and simultaneously engaging said wire terminal and securing said wire terminal on the terminal support.

11. The electrical adapter of claim 9, wherein the terminal support includes a depression receiving a protuberance of said wire terminal and restricting movement of said wire terminal relative to the terminal support to at least an extent.

12. The electrical adapter of claim 11, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to said wire terminal and restricting removal of said protuberance of said wire terminal from the depression of the terminal support.

13. The electrical adapter of claim 12, wherein the depression opens towards the second housing shell and is at least partially aligned with the terminal rib.

14. The electrical adapter of claim 12, wherein the terminal rib and the terminal support prevent the first housing shell and the second housing shell from being connected to one another when said wire terminal is improperly seated on the terminal support.

15. The electrical adapter of claim 4, wherein the second housing shell includes a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to a wire terminal of said wire and restricting movement of said wire terminal relative to the first housing shell to at least an extent.

16. The electrical adapter of claim 4, wherein the second housing shell includes a cable rib projecting toward the first housing shell, the cable rib having a free end contacting and/or disposed in close proximity to said flat cable and restricting movement of said flat cable relative to the first housing shell to at least an extent.

17. The electrical adapter of claim 4, wherein the second housing shell includes a lip projecting toward the wire opening, the lip contacting and/or disposed in close proximity to a free end of a wire terminal of said wire and restricting movement of said wire terminal away from a conductor of said flat cable engaged therewith.

18. The electrical adapter of claim 4, wherein: the female connector includes (i) a pair of tracks and (ii) a tab; and the male connector includes (i) a pair of guide rails slidably engageable with the pair of tracks and (ii) a latch releasably engageable with the tab.

19. An electrical adapter for connecting a flat cable and a wire, comprising: a first housing shell; a second housing shell releasably connected to the first housing shell; an interior space defined by and between the first housing shell and the second housing shell; a wire opening via which said wire extends into the interior space; a cable opening via with said flat cable extends into the interior space; a male connector disposed on the second housing shell outside of the interior space; and a female connector disposed on the first housing shell outside of the interior space; wherein the first housing shell includes: a retention protrusion projecting toward the second housing shell, the retention protrusion disposed in a slot of said flat cable and securing said flat cable to the first housing shell; a terminal support projecting toward the second housing shell and on which a wire terminal of said wire is disposed; and at least one latch disposed adjacent to the terminal support and projecting toward the second housing shell, the at least one latch engaging said wire terminal and securing said wire terminal on the terminal support; wherein the second housing shell includes: a terminal rib projecting toward the first housing shell, the terminal rib having a free end contacting and/or disposed in close proximity to said wire terminal and restricting disengagement of the at least one latch and said wire terminal to at least an extent; a cable rib projecting toward the first housing shell, the cable rib having a free end contacting and/or disposed in close proximity to said flat cable and restricting removal of the retention protrusion from said slot of said flat cable; and a lip projecting from the cable rib toward the terminal rib, the lip contacting and/or disposed in close proximity to a free end of said wire terminal and restricting movement of said wire terminal away from a conductor of said flat cable engaged therewith; and wherein the male connector and the female connector are releasably engageable with one another such that (i) the second housing shell is releasably connectable to another first housing shell of another electrical adapter and (ii) the first housing shell is releasably connectable to another second housing shell of another electrical adapter.

20. A modular adapter assembly for connecting a plurality of flat cables and a plurality of wires, comprising a plurality of electrical adapters according to claim 19, wherein adjacent adapters of the plurality of adapters are releasably connected to one another via engagement of the male connector of a first adapter of the adjacent adapters and the female connector of a second adapter of the adjacent adapters.